No of Pages : 3 Course Code : 23MX11

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester: 1

23MX11 MATHEMATICAL FOUNDATIONS OF COMPUTER SCIENCE

Time: 3 Hours **Maximum Marks: 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, subdivision (b) carries 7 marks each and subdivision (c) carries 15 marks each.
3. Graph sheet is to be provided
4. Chi-Square table/t-table/Z-table are permitted to use
4. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

1. a) In a class, 40% of the students enrolled for Mathematics and 70% enrolled for Computer Science. If 15% of the students enrolled for both Mathematics and Computer Science, what % of the students of the class did not enroll for either of the two subjects?

b) i) Show that the relation R in the set A of all the books in a library of a college, given by R = {(*x*, *y*) : *x* and *y* have same number of pages} is an equivalence relation. [4]

ii) Consider f: N → N, g: N → N and h: N → R defined as follows.

$$f(x) = 2x, \quad g(y) = 3y + 4 \text{ and } h(z) = \sin z, \forall x, y \text{ and } z \text{ in } N.$$

Show that $h\bullet(g\bullet f) = (h\bullet g)\bullet f$. [3]

c) i) If 18486 is divisible by 18, then 18486 is divisible by 9. If 18486 is divisible by 9, then the sum of the digits of 18486 is divisible by 9. Therefore, If 18486 is divisible by 18, then the sum of the digits of 18486 is divisible by 9. Determine the validity of the given arguments. [10]

ii) There are n people in a room, where n is an integer greater than or equal to 1. Each person shakes hands once with every other person. Prove by mathematical induction that the number of handshakes is given by (½) n(n-1). [5]

2. a) Write the dual of the given linear programming problem.

Maximize $Z = 2x1 + 3x2 + x3$
Subject to:

$$4x1 + 3x2 + x3 = 6$$
$$x1 + 2x2 + 5x3 = 4$$
$$x1, x2, x3 \geq 0$$

b) A factory manufactures two products A and B. To manufacture one unit of A, 1.5 machine hours and 2.5 labor hours are required. To manufacture product B, 2.5 machine hours and 1.5 labor hours are required. In a month, 300 machine hours and 240 labor hours are available. Profit per unit for A is Rs. 50 and for B is Rs. 40. Mathematically formulate the given linear programming problem and solve the same using Graphical method.

c) i) A) Solve the following linear programming problem using **Simplex method** [10]

Maximize $12x1 + 18x2 + 10x3$

Subject to:

$$\begin{aligned} 2x1 + 3x2 + 4x3 &\leq 50 \\ x1 - x2 - x3 &\geq 0 \\ x2 - 1.5x3 &\geq 0 \text{ and } x1, x2, x3 \geq 0 \end{aligned}$$

B) Solve the following transportation problem. [5]

Source	Destination				Supply
	D1	D2	D3	D4	
O1	3	1	7	4	300
O2	2	6	5	9	400
O3	8	3	3	2	500
Demand	250	350	400	200	

(OR)

ii) Find the optimal solution of the following linear programming problem using **Big - M method**.

Minimize $Z = 3x1 + 2x2$

Subject to:

$$\begin{aligned} 2x1 + x2 &\geq 10 \\ -3x1 + 2x2 &\leq 6 \\ x1 + x2 &\geq 6 \text{ and } x1, x2 \geq 0 \end{aligned}$$

3. a) Find the probability of getting 53 Sundays in a leap year.

b) i) Suppose there are two bags in a box, which contain the following marbles: Bag 1 contains 7 red and 3 green marbles; Bag 2 contains 2 red and 8 green marbles. If a person randomly select one of the bags and then randomly select one marble from that bag, what is the probability that it is a green marble? [4]

ii) A random variable X has the following probability distribution. Find the value of $P(1.5 < X < 4.5 \mid X > 2)$ [3]

X	0	1	2	3	4	5	6	7
P(x)	0	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{1}{100}$	$\frac{2}{100}$	$\frac{17}{100}$

c) i) A) The temperature in Degree Fahrenheit during the month of June is listed below. What is the Mean, Median and Mode of the temperatures for the month of June? [5]

70, 76, 76, 74, 70, 70, 72, 74, 78, 80, 74, 74, 78, 76, 78, 76, 74, 78, 80, 76, 72, 74, 78, 80, 73, 75, 80, 80, 71, 76.

B) Given the following frequency distribution of weights of 60 apples, calculate the Mode for grouped data. [10]

Weights (grams)	65-84	85-104	105-124	125-144	145-164	165-184	185-204
Frequency	9	10	17	10	5	4	5

(OR)

ii) A) Suppose that you have a fair 4-sided die, and let X be the random variable representing the value of the number rolled. Write down the moment generating function for X. Using this moment generating function, compute the first and second moments of X. [5]

B) If the joint probability density function of a two-dimensional random variable (X, Y) is given by $f(X,Y) = 8xy$ for $0<x<1$ and $0<y<x$; otherwise 0. Find: a) the marginal density functions of X and Y b) the conditional density functions and c) Check for independence of X and Y and $P[Y < \frac{1}{4} \mid X < \frac{1}{2}]$ [10]

4. a) What is the significance of exponential smoothing in time series analysis?

b) A professor wants to know if her introductory statistics class has a good grasp of basic mathematics. Six students are chosen at random from the class and given a mathematics proficiency test. The professor wants the class to be able to score on average above 70 on the test. The six students get the following scores: 62, 92, 75, 68, 83, and 95. Can the professor claim that the mean score for the class on the test would be above 70 with 0.05 significance level?

c) Given the following data which relates to a study made by a retail merchant to determine the relation between his sales and weekly advertising expenditure. The following data was recorded by him for a period of 12 days.

Advertising Cost X	40	20	25	20	30	50	40	20	50	40	25	50
Sales Y	385	400	395	365	475	440	490	420	560	525	480	510

i) Find the correlation coefficient, assess the statistical significance of the value and interpret your results

ii) Find the equation of the regression line to predict weekly sales from the advertising cost and estimate weekly sales if the advertising cost is 35

/END/

FD/RL